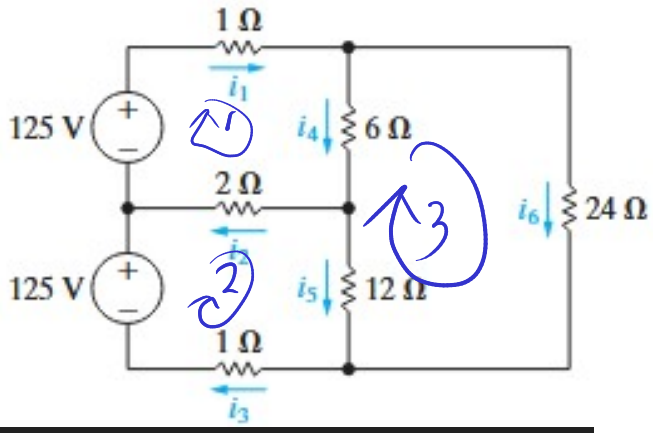
EE 2010
Fall 2026

Name _____

Game 16
9/21/2026

For the following circuit

a) Write the mesh current equations for the three meshes, and solve the system of equations
b) Calculate all currents (i_1-i_6) in the circuit



$$i_1 = I_1$$
$$i_2 = I_1 - I_2$$
$$i_3 = I_2$$
$$i_4 = I_1 - I_3$$
$$i_5 = I_2 - I_3$$
$$i_6 = I_3$$
$$KVL_1 = 125 + I_1 + 6I_1 - 6I_3 + 2I_1 - 2I_2 = 125 + 9I_1 - 2I_2 - 6I_3 = 0$$
$$KVL_2 = 125 - 2I_1 + 2I_2 + 12I_2 - 12I_3 + I_2 = 125 + 15I_2 - 2I_1 - 12I_3 = 0$$
$$KVL_3 = 24I_3 - 6I_1 + 6I_3 - 12I_2 + 12I_3 = 42I_3 - 6I_1 - 12I_2 = 0$$

Using KVL3:

$$6I_1 = 42I_3 - 12I_2$$
$$I_1 = 7I_3 - 2I_2$$
$$KVL_1: \ 125 + 63I_3 - 18I_2 - 2I_2 - 6I_3 = 0$$
$$125 + 57I_3 - 20I_2 = 0$$
$$KVL_2: \ 125 + 15I_2 - 14I_3 + 4I_2 - 12I_3 = 0$$
$$125 + 19I_2 - 26I_3 = 0$$

$COMBINING$ 1&2

$$57I_3 - 20I_2 = 19I_2 - 26I_3$$
$$83I_3 = 39I_2$$
$$I_2 = \frac{83}{39} I_3$$
$$I_1 = \frac{273I_3 - 166I_3}{39} = \frac{107}{39} I_3 \approx 2.74I_3$$
$$-125 = 9\left(\frac{107}{39} I_3\right) - 2\left(\frac{83}{39} I_3\right) - 6I_3$$
$$-125 = \frac{963I_3 - 166I_3 - 234I_3}{39}$$
$$-125 = \frac{563}{39} I_3$$
$$I_3 = -8.66A$$